

Integers, Place Value and Arithmetic

Module 2 · Homework Full Solutions

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Each solution follows the CM six-part standard: main solution · second method · faster contest method · concept tested · likely student mistake · extension question.

1. Answer (B) 80

Retrieval · M1 — contest scoring (5/6/8). In a 25-question contest, questions 1–10 are worth 5 points each, 11–20 are worth 6 points each and 21–25 are worth 8 points each. Priya gets questions 1–10 right and five of the questions 11–20 right. All her other answers are wrong. What is her score?

- **Main solution.** Part A: $10 \times 5 = 50$. Part B: $5 \times 6 = 30$. Wrong answers score 0. Total $50 + 30 = 80$.
- **Second method.** Count all 15 correct at 5 points (75), then add the extra 1 point for each of the 5 Part B questions: $75 + 5 = 80$.
- **Faster contest method.** Part A is worth 50 in total; add $5 \times 6 = 30$.
- **Concept tested.** Contest scoring geometry (Part A/B values)
- **Likely student mistake.** (A) 75 — scored all 15 correct answers at 5 points
- **Extension question.** What is the smallest number of correct answers that scores at least 100? *Answer:* Use the highest values first: five Part C answers (40) and ten Part B answers (60) make 100 with 15 correct answers. 14 answers score at most $40 + 9 \times 6 = 94$, so 15 is the smallest.

2. Answer (D) 400

Retrieval · M1 — try a small case. What is the value of $1 + 3 + 5 + 7 + \dots + 39$?

- **Main solution.** There are 20 odd numbers from 1 to 39. Small cases: $1 = 1$, $1 + 3 = 4$, $1 + 3 + 5 = 9$ — the sum of the first k odd numbers is $k \times k$. So the sum is $20 \times 20 = 400$.
- **Second method.** Pair first and last: $1 + 39 = 40$, $3 + 37 = 40$, ...; 20 terms make 10 pairs, $10 \times 40 = 400$.
- **Faster contest method.** Pair first and last: 20 terms make 10 pairs of 40 $\rightarrow 400$.
- **Concept tested.** Small cases reveal a pattern (sums of odd numbers are squares)
- **Likely student mistake.** (E) 420 — summed $2 + 4 + \dots + 40$ (the evens); (A) 210 — summed $1 + 2 + \dots + 20$
- **Extension question.** What is $1 + 3 + 5 + \dots + 99$? *Answer:* 50 odd numbers $\rightarrow 50 \times 50 = 2\,500$.

3. Answer (B) 16

Retrieval · M1 — work backward. Tomas thinks of a number. He multiplies it by 4 and then subtracts 9. His result is 55. What number did he think of?

- **Main solution.** Undo each step in reverse order. Result $55 \rightarrow$ add back the 9 $\rightarrow 64 \rightarrow$ divide by 4 $\rightarrow 16$.
- **Second method.** Equation: $4x - 9 = 55$, so $4x = 64$ and $x = 16$.
- **Faster contest method.** Undo in reverse: $55 + 9 = 64$, $64 \div 4 = 16$. Check: $16 \times 4 - 9 = 55$.
- **Concept tested.** Working backward with inverse operations
- **Likely student mistake.** (A) 11.5 — undid the steps in the wrong order: $(55 - 9) \div 4$
- **Extension question.** If he had subtracted 9 first and then multiplied by 4, getting 55, what would be his number? *Answer:* $4(x - 9) = 55$ gives $x - 9 = 13.75$, $x = 22.75$.

4. Answer (E) 0.07

Current topic · M2. Which of the following numbers is the smallest?

- **Main solution.** Write every number to four decimal places: 0.7000, 0.0700, 0.7070, 0.0770, 0.0707. The smallest is $0.0700 = 0.07$.
- **Second method.** Compare tenths first: 0.07, 0.077 and 0.0707 have 0 tenths. Hundredths: all 7. Thousandths: 0, 7, 0. Ten-thousandths: 0.0700 vs 0.0707 $\rightarrow$ 0.07 is smallest.
- **Faster contest method.** Pad to four places: 7000, 0700, 7070, 0770, 0707.
- **Concept tested.** Ordering decimals
- **Likely student mistake.** (D) 0.0707 — more zeros and digits looked smaller
- **Extension question.** List all five numbers from largest to smallest. *Answer:* 0.707, 0.7, 0.077, 0.0707, 0.07.

5. Answer (A) 4

Current topic · M2. A rope 4.8 m long is cut completely into pieces 30 cm long. An identical rope is cut completely into pieces 40 cm long. How many more 30 cm pieces are there than 40 cm pieces?

- **Main solution.** $4.8 \text{ m} = 480 \text{ cm}$. $480 \div 30 = 16$ pieces; $480 \div 40 = 12$ pieces. $16 - 12 = 4$.
- **Second method.** Every 120 cm of rope gives 4 pieces of 30 cm but only 3 pieces of 40 cm — one extra piece. 480 cm is 4 lots of 120 cm, so there are 4 extra pieces.
- **Faster contest method.** $480 \div 30 = 16$ and $480 \div 40 = 12$.
- **Concept tested.** Division and comparison of counts (with units)
- **Likely student mistake.** (B) 10 — subtracted the lengths $40 - 30$
- **Extension question.** With a third rope cut into 60 cm pieces, how many fewer pieces than the 30 cm rope? *Answer:* $480 \div 60 = 8$; $16 - 8 = 8$ fewer.

6. Answer (C) \$10.50

Current topic · M2. A parking lot charges \$1.75 for every 15 minutes or part of 15 minutes. A car parks from 9:52 a.m. until 11:20 a.m. How much does it pay?

- **Main solution.** From 9:52 to 10:52 is 60 min; 10:52 to 11:20 is 28 min. Total 88 min. 88 min is more than 5 blocks (75 min) and at most 6 blocks (90 min), so the car pays for 6 blocks: $6 \times \$1.75 = \10.50 .
- **Second method.** Count up in 15-minute blocks from 9:52: 10:07, 10:22, 10:37, 10:52, 11:07, 11:22. The 6th block ends at 11:22, after 11:20 → 6 blocks.
- **Faster contest method.** 9:52 → 11:20 is 1 h 28 min = 88 min; 90 min is 6 blocks; $6 \times 1.75 = 10.50$.
- **Concept tested.** Time difference with regrouping; money $\times$ count; ceiling of a division
- **Likely student mistake.** (A) \$8.75 — ignored the part block (5 blocks for 88 min)
- **Extension question.** What is the latest time the car could leave and still pay \$10.50? *Answer:* The 6th block ends at 11:22 a.m.

7. Answer (C) 8°C

Current topic · M2. At midnight the temperature was -6°C . By 4 a.m. it had fallen 9 degrees. By noon it had risen 23 degrees from its 4 a.m. value. What was the temperature at noon?

- **Main solution.** Midnight -6°C . Fall 9: $-6 - 9 = -15^{\circ}\text{C}$ at 4 a.m. Rise 23: $-15 + 23 = 8^{\circ}\text{C}$ at noon.
- **Second method.** Net change: $-9 + 23 = +14$. Start -6 , so $-6 + 14 = 8$.
- **Faster contest method.** Net change $-9 + 23 = +14$; $-6 + 14 = 8$.
- **Concept tested.** Integer change on a number line
- **Likely student mistake.** (D) 20°C — started from $+6$ instead of -6
- **Extension question.** Which of the three readings was lowest, and how many degrees below the noon reading was it? *Answer:* The 4 a.m. reading, -15°C , was lowest; it was 23 degrees below the noon reading of 8°C .

8. Answer (B) 7:18 a.m.

Interleaved · M1 work backward + M2 time. Maya's bus comes at 8:17 a.m. She needs 18 minutes to get ready, 25 minutes for breakfast and 12 minutes to walk to the stop, and she wants to be at the stop 4 minutes early. What is the latest time she can get up?

- **Main solution.** Add the times needed: $18 + 25 + 12 + 4 = 59$ min. Count back 59 min from 8:17: $8:17 - 17 \text{ min} = 8:00$, then 42 more minutes back = 7:18 a.m.
- **Second method.** 59 min = 1 h – 1 min. $8:17 - 1 \text{ h} = 7:17$; add back 1 min → 7:18 a.m.
- **Faster contest method.** Total 59 min = 1 h – 1 min: $8:17 - 1 \text{ h} = 7:17$, + 1 min = 7:18.
- **Concept tested.** Work backward through time with regrouping

- **Likely student mistake.** (C) 7:22 a.m. — forgot the 4 minutes early
- **Extension question.** If breakfast takes 10 minutes longer, when must she get up?
Answer: 69 min before 8:17 → 7:08 a.m.

9. Answer (E) 95

Interleaved · M1 eliminate options + M2 middle value. Which of the following could be the sum of five consecutive whole numbers?

- **Main solution.** Five consecutive numbers are $m - 2$, $m - 1$, m , $m + 1$, $m + 2$. They add to $5 \times m$, a multiple of 5. Only 95 is a multiple of 5: $17 + 18 + 19 + 20 + 21 = 95$.
- **Second method.** Eliminate by trying: the smallest sums are $0+1+2+3+4 = 10$, then 15, 20, 25, ... — they go up by 5, so the sum must end in 0 or 5.
- **Faster contest method.** Five consecutive numbers sum to $5 \times \text{middle}$, so the sum is a multiple of 5.
- **Concept tested.** Sum of consecutive numbers = count $\times$ middle value; eliminate options
- **Likely student mistake.** (B) 63 or (D) 81 — tried a few sums, found none, and guessed
- **Extension question.** Which of the five options could be the sum of four consecutive whole numbers? *Answer:* Four consecutive numbers k , $k + 1$, $k + 2$, $k + 3$ add to $4k + 6$. Subtract 6 and test for a multiple of 4: only $74 - 6 = 68 = 4 \times 17$ works, so $74 = 17 + 18 + 19 + 20$.

10. Answer (C) 450

Challenge · M2 — Part C. The number N is written with fifty 9s: $N = 999\dots9$. What is the sum of the digits of $7 \times N$?

- **Main solution.** $N = 10^{50} - 1$, so $7 \times N = 7 \times 10^{50} - 7$. Small cases: $7 \times 9 = 63$, $7 \times 99 = 693$, $7 \times 999 = 6\,993$. With n nines the product is 6, then $(n - 1)$ nines, then 3. For $n = 50$: $6 + 49 \times 9 + 3 = 450$.
- **Second method.** Digit-sum fact from the small cases: $63 \rightarrow 9$, $693 \rightarrow 18$, $6\,993 \rightarrow 27$: the digit sum is $9 \times (\text{number of 9s})$. So $9 \times 50 = 450$.
- **Faster contest method.** $7 \times N = 7 \times 10^{50} - 7 = 6$, then forty-nine 9s, then $3 \rightarrow 6 + 49 \times 9 + 3$.
- **Concept tested.** $10^n - 1$ structure; digit sum
- **Likely student mistake.** (A) 350 — assumed the answer is 7×50
- **Extension question.** What is the sum of the digits of $13 \times N$ (N with fifty 9s)?
Answer: $13 \times 999 = 12\,987$ and $13 \times 9\,999 = 129\,987$: the product is 12, then $(n - 2)$ nines, then 87. For $n = 50$: $1 + 2 + 48 \times 9 + 8 + 7 = 450$ (again 9×50).